

Total No. of Questions : 4]

SEAT No. :

P-5143

[Total No. of Pages : 2

[6187]-553

T.E. (Computer Engineering) (Insem.)
DATA SCIENCE AND VISUALIZATION (Honors)
(2019 Pattern) (Semester - I) (310501)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Elaborate different stages of data science process in detail. **[6]**

b) What is supervised Machine Learning? Elaborate with suitable example and enlist its applications. **[5]**

c) Define the terms : **[4]**

- i) Data science
- ii) Big Data

OR

Q2) a) What is Unsupervised Machine Learning? Elaborate with suitable example. **[6]**

b) How is Machine Learning related to Data Science? Explain in detail. **[5]**

c) Define the terms:

- i) Correlation
- ii) Variance **[4]**

P.T.O.

- Q3)** a) Calculate the mean, median, mode of the following data and comment on Skewness of data. [6]
17, 16, 25, 23, 22, 23, 28, 25, 25
- b) Explain normal distribution and its characteristics. [5]
- c) Write a note on conditional Probability. [4]

OR

- Q4)** a) Define Central Tendencies (Mean, Median and Mode) with examples. [6]
- b) What is Bayes Theorem? How it is used to solve classification problem in machine learning? Illustrate with suitable example. [5]
- c) Define the terms related to data science. [4]
- i) Co-Variance
 - ii) Standard Deviation
